

2-Mark Questions & Answers — Data Structures Module 5: Graphs

Q1. Define a Graph and state the Handshaking Lemma.

A **Graph** is $G = (V, E)$ consisting of a set of vertices V and edges E .

- **Handshaking Lemma:** In any undirected graph, the sum of degrees of all vertices is twice the number of edges:

$$\text{Sum of Deg}(v) = 2 * |E|$$

Q2. Differentiate between Directed and Undirected graphs.

- **Undirected Graph:** Edges are bidirectional and symmetric ($(u, v) = (v, u)$).
 - **Directed Graph (Digraph):** Edges have a specific direction ($(u, v) \neq (v, u)$).
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Q3. What is Adjacency Matrix vs Adjacency List?

- **Adjacency Matrix:** 2D array of size $V \times V$, $O(V^2)$ space, $O(1)$ edge query.
 - **Adjacency List:** Array of linked lists, $O(V + E)$ space, ideal for sparse graphs.
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Q4. Define Topological Sorting.

Topological Sorting of a Directed Acyclic Graph (DAG) is a linear ordering of vertices such that for every directed edge (u, v) , vertex u appears before v in the ordering.
